

Akshay Raj As

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Professional Summary

Results-driven Post/Pre-silicon Validation Associate with over 3 years of experience at UST, including 1.5 years as a Team Lead. Specialized in BIOS security features and firmware integration (IFWI/BIOS/ACM). Expertise in executing detailed pre and Post silicon validation tests, debugging, and analyzing issues while contributing to hands-on testing processes. Adept at leading teams and working in high-priority environments, ensuring system stability and security compliance.

Education

Bachelor of Technology (B.Tech) in Electronics and Communication Engineering
Trinity College of Engineering, Naruvamoodu, Trivandrum — 2020
CGPA: 7.88

Higher Secondary Education
Christ Nagar EM HSS, Kowdiar — 2016
Stream: BioMaths
Percentage: 81%

Internship Experience

Intern
ISRO — 2020

- Worked on a CCSDS Compliant LDPC Encoder as part of the project team.
- Contributed to the design and testing phases, gaining hands-on experience in coding and debugging.

Professional Experience

Post/Pre-silicon Validation Associate / Team Lead
UST (for Intel) —27/07/2021 to Present

- Led a team in testing BIOS security features such as TXT, TDX, MKTME, Boot Guard, and BIOS Guard, ensuring the robustness of Intel's latest BIOS releases.
- Executed pre-written test cases in a virtual environment using Simics, ensuring high-fidelity simulations for thorough validation of BIOS and firmware releases.
- Conducted firmware integration testing for IFWI/BIOS/ACM, ensuring system stability, security, and functionality across platforms.

- Performed critical sanity and functional testing for BIOS and ACM firmwares, identifying potential issues early in the development cycle.
 - Tested and validated 14th Gen Intel processors and upcoming releases for both basic acceptance and functional testing.
 - Applied root cause analysis using debug log parsing to diagnose and resolve issues efficiently.
 - Actively engaged with developers, clients, and internal teams to discuss defect resolutions and improve testing procedures.
 - Led test case assignment, regression issue identification, and provided solutions to improve test effectiveness.
 - Developed detailed post-test reports, providing actionable insights and recommendations to the team.
 - Worked closely with the team to refine testing methodologies and improve overall testing efficiency.
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Technical Skills

- BIOS Security Testing: Expertise in testing BIOS security features including TXT, TDX, MKTME, Boot Guard, and BIOS Guard.
 - Firmware Integration Testing: Proficient in IFWI/BIOS/ACM integration testing.
 - Manual Testing & Debugging: Skilled in BIOS firmware troubleshooting and defect resolution.
 - Intel Platform Testing: Experience in power management testing and validation of various Intel platforms.
 - Test Case Development & Validation: Strong knowledge of creating and executing test cases for BIOS security and firmware.
 - Programming Knowledge: Basic proficiency in Python and C++ for code editing and understanding.
 - Tools: Chipsec, HSD internal tools, and Simics for high-fidelity simulations.
 - Operating Systems: Comfortable working in Linux environments (command line proficiency).
 - Defect Tracking: Proficient in bug lifecycle management and defect tracking.
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Certifications

- Cambridge English Entry Level Certificate in ESOL International
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Personal Strengths and Interests

- Strong analytical, problem-solving, and debugging skills.
 - Excellent communication and teamwork abilities with a focus on delivering high-quality results.
 - Detail-oriented, with a passion for continuous learning and improving work processes.
 - Adaptable to changing environments, challenges, and team dynamics.
 - Keen interest in emerging semiconductor technologies and advancements in gaming.
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Declaration

I hereby declare that the information provided above is true and accurate to the best of my knowledge.

Akshay Raj As
26/10/2025
